THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 1

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The region around a magnet in which its force can be felt is the 1

(a) gravitational field **(b)** magnetic field
(c) light field **(d)** electric field

2. A current-carrying wire produces 1

(a) no field **(b)** only light **(c)** only sound **(d)** a magnetic field

3. The force on a current-carrying conductor is shown by 1

(a) Fleming's right-hand rule **(b)** Fleming's left-hand rule
(c) Snell's law **(d)** Joule's law

4. The SI unit for the strength of the magnetic field is the 1

(a) ampere **(b)** henry **(c)** weber only **(d)** tesla

5. Oersted observed that a compass needle moves near a wire carrying 1

(a) sound **(b)** no current
(c) light **(d)** an electric current

6. In Fleming's left-hand rule, the forefinger indicates 1

(a) the field **(b)** the voltage **(c)** the force **(d)** the current

7. Like poles of magnets 1

(a) merge **(b)** do not interact
(c) attract each other **(d)** repel each other

8. When the current in a wire is reversed, the compass needle deflects 1

(a) not at all **(b)** in the same direction
(c) randomly **(d)** in the opposite direction

9. In Fleming's left-hand rule, the thumb indicates 1

(a) current **(b)** resistance **(c)** force or motion **(d)** field

10. Unlike poles of magnets 1

(a) attract each other **(b)** do not interact
(c) repel each other **(d)** melt

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is a magnetic field? 2

12. State the right-hand thumb rule. 2

13. Who discovered electromagnetic induction? 2

14. Name the poles of a magnet. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Draw (describe) the field lines of a bar magnet and mention the direction. 3

16. State Fleming's left-hand rule. 3

17. Name the three wires in a domestic circuit with their colours. 3

OR

State any three properties of magnetic field lines. 3

18. State Fleming's right-hand rule. 3

OR

Why is a fuse connected in the live wire? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the properties of magnetic field lines. 5

OR

Describe the principle, construction and working of an AC generator. 5

20. Describe Oersted's experiment with a diagram description. State its conclusion. 5

OR

Distinguish between AC and DC giving the sources of each. State one advantage of AC. 5

END OF QUESTION PAPER — Total: 40 Marks

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